Precept 5

Perfect Competition

1. Consider a market of infinitesimally small firms, where the firm with index $a \geq 0$ has cost function $C_a(q) = \frac{a^2}{2} + \frac{q^2}{2}$. To compute market supply, use the fact that for every $m > 0$, there is a mass m of firms with indices in $[0, m]$. (For practical purposes, just take this to mean that there are m firms with indices in $[0, m]$ and don't worry about the possibility of a non-integer quantity). Firms are price takers and face demand curve $Q^D(p) = 10 - 3p$.

   (a) Find the short-run shutdown prices for a firm with index a, assume that fixed costs are unavoidable.

   Answer: in the short run, a firm shuts down if $p < \min AVC$. Here, $\frac{a^2}{2}$ is a fixed cost (doesn't depend on production choices) so variable costs are $\frac{q^2}{2}$; thus $AVC = \frac{q}{2}$, which clearly reaches a minimum, at $q = 0$, of zero. Thus the shutdown price is zero. (We did the long-run version in class, finding shutdown price $p = \min AC = a$, achieved by producing $q = a$: this is how to set $MC = AC$, which we learned was the recipe for minimizing AC).

   (b) Find the SR individual supply curve.

   Answer: Since $MC = q$, which is increasing in q, the SOC holds. So a firm produces where the FOC $p = MC$ holds, so long as p exceeds the shutdown price. Since $MC = q$, this yields $p = q$. Thus each firm's SR supply curve is $q(p) = p$, so long as $p \geq 0$.

2. Calculate the long-run cost function for a firm that produces according to production technology $f(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$ where K is capital and L is labor.

   Answer: this asks for the minimum cost of producing q units. The bundle that does so most efficiently solves:

   - efficient input spending: here, analogous to the "optimal \$ allocation" condition for consumer theory, it is to equate bang-per-buck across goods, but here the bang-per-buck is marginal product divided by price, rather than marginal utility over price. So: s

$$\begin{aligned}\frac{MP_K}{MP_L} &= \frac{r}{w} \Rightarrow \frac{\frac{1}{2}K^{-\frac{1}{2}}L^{\frac{1}{2}}}{\frac{1}{2}K^{\frac{1}{2}}L^{-\frac{1}{2}}} = \frac{r}{w} \\ &\Rightarrow \frac{L}{K} = \frac{r}{w}, \text{ or } wL = rK\end{aligned}$$

   - actually produce q units: this asks that $K^{\frac{1}{2}} L^{\frac{1}{2}} = q$
   - plugging the efficient spending condition (namely $L = \frac{r}{w}K$) into the production constraint, we get

$$\begin{aligned}K^{\frac{1}{2}}\left(\frac{r}{w}K\right)^{\frac{1}{2}} &= q \Rightarrow K\sqrt{\frac{r}{w}} = q \Rightarrow K = q\sqrt{\frac{w}{r}} \\ &\Rightarrow L = \frac{r}{w}K = \frac{r}{w}\left(q\sqrt{\frac{w}{r}}\right) = q\sqrt{\frac{r}{w}}\end{aligned}$$

- Now, finally, the cost of producing q is the cost of the input bundle that most efficiently produces q. We just figured out that this bundle is $(K^*, L^*) = \left(q\sqrt{\frac{w}{r}}, q\sqrt{\frac{r}{w}}\right)$. So,

$$C^{LR}(q) = r\left(q\sqrt{\frac{w}{r}}\right) + w\left(q\sqrt{\frac{r}{w}}\right) = 2\sqrt{rw}\cdot q$$

Double Auctions and Externalities

(We haven't learned externalities but this should still be doable; we'll discuss them late in the semester).

3. There are three local Christmas tree farms, each with up to 2 trees for sale. Farm #1 incurs a cost of \$90 per tree it cuts down, Farm #2's cost is \$50 per tree, and Farm #3's cost is \$70 per tree. And there are 12 buyers looking for Christmas trees: odd-numbered students 1,3,5,7,9,11 have valuations equal to 20 times their student number, and even-numbered students have valuation zero.

   (a) Find all possible equilibrium prices and quantities.

   Answer: ignoring the buyers with valuation zero, the payoff matrix, ordering valuations high-low and costs low-high, is:

buyers	220	180	140	100	60	20
sellers	50	50	70	70	90	90

   Clearing the market left to right, we get $q^* = 4$ (the first four pairs should trade, last two should not). For the price, we need to encourage the 4th trade, so $70 \le p^* \le 100$, and discourage the 5th, so $60 \le p^* \le 90$: the binding inequalities are $70 \le p^* \le 90$, so any price in this range works.

   (b) Now suppose that each tree purchased generates a \$40 positive externality for the buyer's neighborhood, as the neighbors feel happy when they see trees in windows. Calculate the socially optimal number of Christmas tree sales, i.e. that maximizes social value minus social cost.

   Answer: this raises the social value of each sale to \$40 above the buyer's, so we get

social values	260	220	180	140	100	60
costs	50	50	70	70	90	90

   The social value exceeds the cost for the first 5 units but not the 6th, and so the socially optimal # trees sold is 5.

   (c) As in (c), each tree purchased generates a \$40 positive externality for the neighborhood. Accordingly, a neighborhood planner decides to encourage tree sales buy offering a subsidy of $\$s$ to anyone who purchases a tree. Find all subsidies $\$s$ that will lead to the socially optimal number of tree sales, as in (c).

   Answer: going back to the valuation matrix from (a), let's add a subsidy s to each buyer value (he gets s more from purchasing than he did before, thanks to the subsidy):

buyers	$220+s$	$180+s$	$140+s$	$100+s$	$60+s$	$20+s$
sellers	50	50	70	70	90	90

   We want a subsidy that makes the 5th trade happen but not the 6th: so we need $60 + s \ge 90 \Leftrightarrow s \ge 30$, and $20 + s \le 90 \Leftrightarrow s \le 70$. I.e. a subsidy between \$30 and \$70.